

i) Check whether the foll. series are absolutely or conditionally convergent.

a) $\sum_{n=1}^{\infty} (-1)^{n-1} [\sqrt{n+1} - \sqrt{n}]$

$$|u_n| = |(-1)^{n-1} (\sqrt{n+1} - \sqrt{n})| = \sqrt{n+1} - \sqrt{n}$$
$$= \sqrt{n+1} - \sqrt{n} \times \frac{\sqrt{n+1} + \sqrt{n}}{\sqrt{n+1} + \sqrt{n}}$$

$$|u_n| = \frac{(n+1) - n}{\sqrt{n+1} + \sqrt{n}} = \frac{1}{\sqrt{n+1} + \sqrt{n}} = \frac{1}{\sqrt{n(1+\frac{1}{n})} + \sqrt{n}}$$

$$\therefore |u_n| = \frac{1}{\sqrt{n} \left(\sqrt{1+\frac{1}{n}} + 1 \right)}$$

$$\text{let } v_n = \frac{1}{\sqrt{n}}$$

$$\lim_{n \rightarrow \infty} \frac{|u_n|}{v_n} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{1+\frac{1}{n}} + 1} = \frac{1}{\sqrt{1+0} + 1} = \frac{1}{2} \neq 0 \text{ \& finite}$$

$$\sum v_n = \sum \frac{1}{\sqrt{n}} = \sum \frac{1}{n^{\frac{1}{2}}} \text{ is a p-series with } p = \frac{1}{2} < 1$$

$\therefore \sum v_n$ diverges.

$\therefore$ By limit form of comparison test, $\sum |u_n|$ also diverges.
($\therefore \sum u_n$ is not absolutely convergent)

Now, $\sum (-1)^{n-1} (\sqrt{n+1} - \sqrt{n})$ is alternating with
 $u_n = \sqrt{n+1} - \sqrt{n}$

$$U_{n-1} = \sqrt{n} - \sqrt{n-1}$$

$$U_n - U_{n-1} = \sqrt{n+1} - \sqrt{n} - \sqrt{n} + \sqrt{n-1}$$

$$= \sqrt{n+1} - 2\sqrt{n} + \sqrt{n-1}$$

$$< \sqrt{n+1} - 2\sqrt{n+1} + \sqrt{n+1} = 0$$

$$\therefore U_n - U_{n-1} < 0 \quad \text{for } n > 1.$$

$$u_n = \sqrt{n+1} - \sqrt{n} \times \frac{\sqrt{n+1} + \sqrt{n}}{\sqrt{n+1} + \sqrt{n}} = \frac{(n+1) - n}{\sqrt{n+1} + \sqrt{n}} = \frac{1}{\sqrt{n+1} + \sqrt{n}}$$

$$u_n = \frac{1}{\sqrt{n} \left(\sqrt{1 + \frac{1}{n}} + 1 \right)}$$

$$\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n} \left(\sqrt{1 + \frac{1}{n}} + 1 \right)} = 0$$

$\therefore$ By Leibnitz's test, $\sum (-1)^{n-1} (\sqrt{n+1} - \sqrt{n})$ converges

$\therefore \sum (-1)^{n-1} [\sqrt{n+1} - \sqrt{n}]$ converges conditionally.

b) $\sum \frac{(-1)^{n-1} n}{n^2+1} = \sum U_n$

$$|U_n| = \frac{n}{n^2+1} = \frac{n}{n^2(1+\frac{1}{n^2})} = \frac{1}{n(1+\frac{1}{n^2})}$$

$$V_n = \frac{1}{n}$$

$$\lim_{n \rightarrow \infty} \frac{|U_n|}{V_n} = \lim_{n \rightarrow \infty} \frac{1}{1+\frac{1}{n^2}} = 1 \neq 0 \text{ \& finite}$$

$\sum V_n = \sum \frac{1}{n}$ is a p-series with $p=1$

$\therefore \sum V_n$ diverges

$\therefore$ By Comparison Test, $\sum |U_n|$ also diverges.

($\because \sum U_n$ is not absolutely convergent)

$\sum \frac{(-1)^{n-1} n}{n^2+1}$ is alternating with $U_n = \frac{n}{n^2+1}$

$$U_{n-1} = \frac{n-1}{(n-1)^2+1}$$

$$U_n - U_{n-1} = \frac{n}{n^2+1} - \frac{n-1}{(n-1)^2+1}$$

$$= \frac{n[(n-1)^2+1] - (n-1)(n^2+1)}{(n^2+1)[(n-1)^2+1]}$$

$$= \frac{n[n^2-2n+2] - [n^3+n-n^2-1]}{(n^2+1)(n-1)^2+1}$$

$$= \frac{n^3-2n^2+2n-n^3-n+n^2+1}{(n^2+1)[(n-1)^2+1]}$$

$$= \frac{-n^2+n+1}{(n^2+1)[(n-1)^2+1]} < 0 \text{ for } n > 1$$

$$\lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} \frac{n}{n^2+1} = \lim_{n \rightarrow \infty} \frac{1}{n(1+\frac{1}{n^2})} = 0$$

$\therefore$ By Leibnitz's test, $\sum \frac{(-1)^{n-1} n}{n^2+1}$ converges

$\therefore \sum \frac{(-1)^{n-1} n}{n^2+1}$ converges conditionally

2) Find interval of convergence of foll. power series.

$$\frac{1}{\sqrt{2}}x + \frac{\sqrt{2}}{\sqrt{3}}x^2 + \frac{\sqrt{3}}{\sqrt{4}}x^3 + \dots$$

$$\rightarrow U_n = \frac{\sqrt{n}}{\sqrt{n+1}} x^n \quad \therefore U_{n+1} = \frac{\sqrt{n+1}}{\sqrt{n+2}} x^{n+1}$$

$$\left| \frac{U_{n+1}}{U_n} \right| = \left| \frac{\sqrt{n+1}}{\sqrt{n+2}} x^{n+1} \times \frac{\sqrt{n+1}}{\sqrt{n}} x^n \right|$$

$$= \left| \frac{(n+1) \cdot x}{\sqrt{n} \sqrt{n+2}} \right| = |x| \cdot \frac{(n+1)}{\sqrt{n} \cdot \sqrt{n} \sqrt{1+\frac{2}{n}}}$$

$$= |x| \cdot \frac{n(1+\frac{1}{n})}{n \sqrt{1+\frac{2}{n}}} = |x| \cdot \frac{(1+\frac{1}{n})}{\sqrt{1+\frac{2}{n}}}$$

$$\lim_{n \rightarrow \infty} \left| \frac{U_{n+1}}{U_n} \right| = \lim_{n \rightarrow \infty} |x| \cdot \frac{(1+\frac{1}{n})}{\sqrt{1+\frac{2}{n}}} = |x|.$$

By Ratio test, power series converges for $|x| < 1$
ie $-1 < x < 1$.

For $x = -1$, series is $-\frac{1}{\sqrt{2}} + \frac{\sqrt{2}}{\sqrt{3}} - \frac{\sqrt{3}}{\sqrt{4}} + \dots$

which is alternating with $U_n = \frac{\sqrt{n}}{\sqrt{n+1}} \therefore U_{n-1} = \frac{\sqrt{n-1}}{\sqrt{n}}$

$$\therefore U_n - U_{n-1} = \frac{\sqrt{n}}{\sqrt{n+1}} - \frac{\sqrt{n-1}}{\sqrt{n}}$$

$$= \frac{n - \sqrt{n-1}\sqrt{n+1}}{\sqrt{n+1} \cdot \sqrt{n}} < 0 \text{ for } n > 1$$

$$\lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} \frac{\sqrt{n}}{\sqrt{n+1}} = \lim_{n \rightarrow \infty} \frac{\sqrt{n}}{\sqrt{n} \sqrt{1+\frac{1}{n}}} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{1+\frac{1}{n}}}$$

$$= \frac{1}{\sqrt{1+0}} = 1 \neq 0$$

$\therefore$ By Leibnitz's test, series oscillates at $x = -1$.

For $x=1$, series is $\sum \frac{\sqrt{n}}{\sqrt{n+1}}$.

$$\therefore U_n = \frac{\sqrt{n}}{\sqrt{n+1}} = \frac{\sqrt{n}}{\sqrt{n} \sqrt{1+\frac{1}{n}}} = \frac{1}{\sqrt{1+\frac{1}{n}}}$$

$$\lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{1+\frac{1}{n}}} = 1 \neq 0$$

$\therefore$ By zero test, series diverges at $x=1$

$\therefore$ Interval of convergence is $-1 < x < 1$.